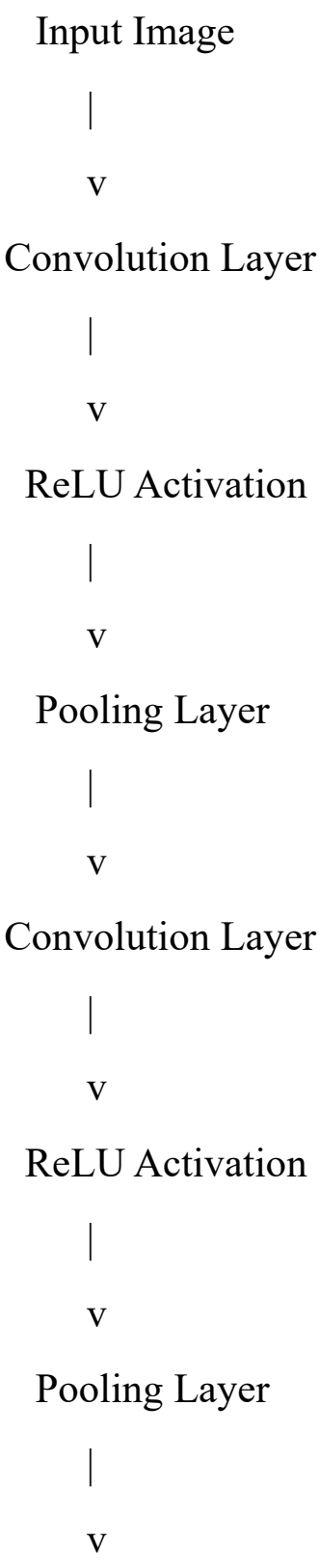
## Module 7: Complete CNN Architecture

### 1. Complete working of CNN

A traditional image-classification CNN follows this general flow:



More Convolution

|

v

Feature Aggregation

|

v

Fully Connected Layer

|

v

Output

The exact architecture depends on the problem. Not every CNN uses this exact sequence.

**2. Role of each component**

**Input image**

The image is represented using numerical pixel values.

**Convolution**

Filters scan the image or previous feature maps and generate feature maps.

**ReLU activation**

ReLU introduces non-linearity.

Formula:

ReLU(x) = max(0, x)

Negative values become zero, while positive values remain unchanged.

**Pooling**

Pooling or another downsampling operation reduces spatial dimensions.

**Deeper convolutional layers**

Deeper layers learn combinations of patterns from earlier feature maps.

**Feature aggregation**

Flattening or global average pooling prepares the extracted features for the classification head.

**Fully connected layer**

The classification head combines the extracted features to produce class scores.

**Output**

The model returns predictions, such as cat or dog.

**3. How does a CNN learn?**

CNN training involves these steps:

Training Image + Correct Label

|

v

Forward Pass

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v

Model Prediction

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v

Loss Function

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v

Backpropagation

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v

Optimizer Updates Weights

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v

Repeat Training

The loss function measures how different the prediction is from the correct target.

Backpropagation calculates gradients. An optimizer uses these gradients to update trainable weights, including convolution filters.

**4. Classification versus detection**

A classification CNN predicts a class for an image.

An object detection model predicts both object classes and locations, commonly using bounding boxes.

A segmentation model predicts classes for pixels or image regions.

Therefore, a basic CNN with a fully connected output is not automatically a complete object detection system.

**5. Common mistakes**

- CNN does not simply merge all edges.
- Convolution performs element-wise multiplication followed by summation.
- Padding does not guarantee equal usage of every pixel.

- Stride reduces spatial resolution, not directly the original image quality.
- Pooling and convolution are separate operations.
- Flattening is not required in every CNN.
- CNN filters are generally learned during training.
- A CNN used for classification needs additional components for object detection.

**Conclusion**

CNN learns visual representations through convolutional operations and uses those representations to solve tasks such as image classification. Training adjusts the filters and other parameters to reduce prediction error.